

Development effects of electrification in Brazil

Motivation, data, and descriptive statistics: Replication of
Lipscomb et al (2013)

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Motivation and prior literature

The authors note a renewed interest of funneling development funds to large-scale infrastructure projects, despite little causal evidence to their effects.

- The key empirical challenge: infrastructure enlargement happens in a **planned** manner, leading to endogeneity in the form of *reverse causality* and *program placement bias*.

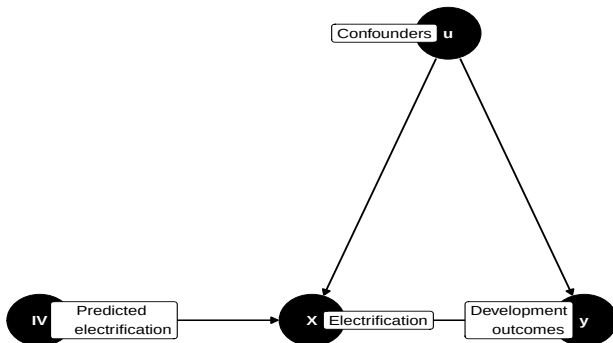
There is overall little discussion in the paper regarding prior literature.

- Previous studies have looked at irrigation dams, highways, and railroads -> **gap #1**: electrification infrastructure.
- Literature has focused on the *macro growth effects* of infrastructure -> **gap #2**: development at a micro-level as DV.

Instrumental variable research design

Given the endogeneity between infrastructure development and development outcomes, a simple OLS regression would not hold.

- Instead, an instrumental variable approach is more appropriate. Therefore, the authors built an IV of *predicted electricity infrastructure*, based on the geographical suitability of hydropower dams.



Data

The paper uses three main data sources to construct a county-level panel dataset (1960-2000) that includes the following:

Historical electricity infrastructure

This data had to be constructed by the authors through data provided by regional electricity companies (feasibility studies, inventories, maps).

Predicted electricity infrastructure

The authors construct a model that uses national budget data and GIS (topographic data). Some assumptions are benchmarked against inventory data of existing electricity infrastructure.

Development indicators and confounders

Census data from the Brazilian Institute of Statistics and Geography (IBGE) and aggregations from census and other surveys by Brazilian Institute for Applied Economic Research (IPEA).

Summary statistics (1/2)

	count	mean	sd	min	max
Electricity	8,730.00	0.72	0.42	0.00	1.00
Percent electrified	8,730.00	0.60	0.33	0.00	1.00
Modeled electricity instrument	8,730.00	0.59	0.47	0.00	1.00
Measures of demand					
Population density	8,730.00	62.71	292.19	0.04	10,097.86
Summary measures of development					
Average value of housing (thousands)	8,730.00	13.05	8.37	0.43	62.53
Human development indicator (index)	8,730.00	0.56	0.17	0.15	0.89
HDI components					
HDI longevity	8,730.00	0.57	0.12	0.17	0.88
HDI salaries	8,730.00	0.47	0.29	0.01	1.13
HDI education	8,730.00	0.51	0.15	0.07	0.88
Employment					
Percent economically active	8,730.00	0.36	0.07	0.17	0.63
Percent employed	8,730.00	0.35	0.06	0.13	0.61
Urban employment	8,730.00	0.34	0.07	0.10	0.61
Rural employment	8,685.00	0.35	0.07	0.00	0.74
Human capital					
Years of schooling	8,730.00	2.77	1.55	0.00	9.61
Illiteracy	8,730.00	32.00	17.96	1.80	89.90
Salaries per capita	8,730.00	0.11	0.08	0.00	0.79
Human capital per capita	6,549.00	19.06	7.12	6.57	59.01

Summary statistics (2/2)

	count	mean	sd	min	max
Population changes					
New migrants in past five years	4,366.00	0.07	0.04	0.00	0.33
Life expectancy	8,730.00	60.10	7.81	38.40	76.92
Population density	8,730.00	78.50	372.91	0.04	11,732.17
Percent of population in urban areas	8,730.00	0.52	0.24	0.02	1.00
Poverty and inequality					
Poverty	8,730.00	60.47	24.92	3.81	99.88
Theil index	8,730.00	0.46	0.13	0.10	1.26
Infant mortality	8,730.00	71.96	50.63	6.93	303.66
Less than four years of education	8,730.00	65.25	21.16	9.07	99.80
Other infrastructure					
Percent of households with running water	8,730.00	0.39	0.28	0.00	0.99
Percent of households with sanitation	8,730.00	0.19	0.27	0.00	0.98
Percent of households with cars	4,366.00	0.01	0.01	0.00	0.13
Geography					
Landslope	8,730.00	1.64	2.01	0.00	21.76

Note: variables for *GDP per capita* and *Industrial GDP per capita* were not replicated, due to them missing from the replication files.

- These variables are not used at any stage of analysis, so not a big issue.